

1. Primary Dataset A: Unstructured Review Data (for ABSA)

- **Dataset Name:** Yelp Open Dataset (`yelp_academic_dataset_review.json`)
- **Official Download Link:** [Yelp Open Dataset](#)
- **Alternative TensorFlow/Fast.ai Mirror (Lighter CSV version):** [Yelp Polarity Reviews on TensorFlow Datasets](#)
- **Target File:** `yelp_academic_dataset_review.json` (or any formatted JSON/CSV with review text, business ID, and date).
- **Where to place it:** Save it directly in your project root directory where `01_ABSA_Extraction.py` is located.

2. Primary Dataset B: Quantitative Daily Demand Data

- **Dataset Name:** Kaggle Store Item Demand Forecasting Dataset
- **Official Download Link:** [Kaggle Store Item Demand Forecasting Challenge](#)
- **Alternative Mirror:** [Kaggle Store Item Demand Forecasting Dataset \(2019–2023\)](#)
- **Target File:** `train.csv` (contains columns: date, store, item, sales).
- **Where to place it:** Save it in your working directory alongside `02_Forecasting_and_SHAP.py`.

3. External Auxiliary Features (Handled Automatically via Code APIs)

1. **Weather Metrics (Temperature & Precipitation):**
 - Handled dynamically via the [Open-Meteo Historical Weather API](#).
2. **National Holidays:**
 - Handled automatically in Python using the `holidays` package (`pip install holidays`).

Quick Setup Summary for Google Colab

If you are running in **Google Colab**, you can download both datasets directly into your runtime via the Kaggle API / Direct Download:

Bash

```
# 1. Install Kaggle API
!pip install -q kaggle

# 2. Download Kaggle Store Demand Dataset
!kaggle competitions download -c demand-forecasting-kernels-only
!unzip -q demand-forecasting-kernels-only.zip

# 3. Download / Load Yelp Review Sample Data
# (Or use the built-in fallback generator in Script 01 if working on a quick prototype)
```

(Note: Script `01_ABSA_Extraction.py` and Script `02_Forecasting_and_SHAP.py` include built-in fallback dataset generators that create mock test data if these files are not present in your folder, allowing you to run and verify the entire pipeline immediately).